

Meta-Forensics Aggregation Audit

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1 Scope

This report aggregates category-level forensic summaries into an overall anomaly score.

2 Load Outputs

```
repo_root <- normalizePath("../", winslash = "/", mustWork = TRUE)
study_id <- Sys.getenv("FORENSICS_STUDY_ID", "lungtime")
report_dir <- file.path(repo_root, "reports", "meta", study_id)

required <- c("meta_category_scores_out.csv", "meta_overall_summary.csv")
missing <- required[!file.exists(file.path(report_dir, required))]
if (length(missing) > 0) {
  stop("Missing files: ", paste(missing, collapse = ", "))
}
```

```
scores <- read.csv(file.path(report_dir, "meta_category_scores_out.csv"), check.names = FALSE)
overall <- read.csv(file.path(report_dir, "meta_overall_summary.csv"), check.names = FALSE)
```

3 Interpretation Key

- **anomaly_score** (category): normalized burden score where higher means more concern within that category.
- **overall_score**: weighted combination of category anomaly scores.
- **risk_tier**: coarse aggregation label (**low**, **moderate**, **high**, **insufficient_data**).

```
knitr::kable(overall, digits = 4)
```

overall_score	risk_tier	n_categories
0.0315	low	4

4 Category Contribution Table

```
knitr::kable(scores, digits = 4)
```

category	metric	raw_value	anomaly_score	weight
randomization	fisher_recalc	0.9004	0.0996	1.0
numeric	median_abs_percent_delta	0.0276	0.0138	1.0
registration	mismatch_rate	0.0000	0.0000	0.8
visual	near_duplicate_rate	0.0000	0.0000	0.8

5 Category Score Plot

```
barplot(
  height = scores$anomaly_score,
  names.arg = scores$category,
  col = "gray75",
  border = NA,
```

```
ylim = c(0, 1),  
main = "Category Anomaly Scores",  
ylab = "anomaly_score"  
)  
abline(h = c(0.2, 0.45), col = "red", lty = 2)
```

